Muhammad Sohel Rana

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Research Interests

Machine Learning, Optimal Transport (OT), Unbalanced Optimal Transport (UOT), Dimensionality Reduction, Neighborhood Embeddings, Scalable Learning Methods, Data Science

Education

Ph.D. in Mathematics (Data Science Track) 2026
The University of Texas at Arlington, USA
Advisor: Dr. Keaton Hamm
Research: Machine Learning, Optimal Transport, Unbalanced Optimal Transport, Dimensionality Reduction

M.S. in Mathematics 2017
Western Kentucky University, USA

M.S. in Applied Mathematics 2008
University of Dhaka, Bangladesh

Experience

Assistant Professor of Mathematics (Tenure-Track)
University of North Texas at Dallas, USA — Starting Fall 2026

Graduate Research Assistant
University of Texas at Arlington, USA — June 2023 – Present

- Conducted research in data science and machine learning with a focus on optimal transport (OT) and unbalanced optimal transport (UOT).
- Developed scalable methods for neighborhood embeddings, dimensionality reduction, and distance-based learning.
- Mentored undergraduate and master's students in collaborative research projects.

Graduate Teaching Assistant
University of Texas at Arlington, USA — Sept. 2019 – May 2021, Sept. 2022 – May 2023

- Led recitation and problem-solving sessions for Calculus and Statistics courses.
- Held office hours and supported students through mentoring and individualized instruction.
- Managed grading and instructional activities using Canvas and Blackboard.

Lecturer
Department of Mathematics and Physics
North South University, Bangladesh — Jan. 2022 – Aug. 2022

- Taught precalculus and Calculus and Analytic Geometry I to over 360 students across multiple sections.

Graduate Teaching Assistant
Western Kentucky University, USA — Aug. 2015 – Dec. 2015

Technical Skills

Languages: Python, LaTeX, Mathematica, C, Fortran
Tools: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Keras, NLTK, Matplotlib, PostgreSQL
Expertise: Machine Learning, Optimal Transport, Unbalanced Optimal Transport, Computer Vision, NLP, Supervised and Unsupervised Learning

Publications

- Muhammad Rana, Abiy Tasissa, HanQin Cai, Yakov Gavriyelov, Keaton Hamm. *Recovering Wasserstein Distance Matrices from Few Measurements.* IEEE ICASSP, 2026.
- Zahidur Talukder, Muhammad Rana, Keaton Hamm, Mohammad A. Islam. *Empowering Clients: Self-Adaptive Federated Learning for Data Quality Challenges.* IEEE EDGE, 2025.

Academic Services

Volunteering Experience

- UTA Calculus Bowl Volunteer (2020, 2021, 2023, 2024)

Peer Review Activities

- Peer Reviewer, American Journal of Undergraduate Research (AJUR), 2025

Recognition

- Mathematics Academic Excellence Scholarship, UTA, 2025
- Glenn and Virginia Powers Memorial Scholarship, Western Kentucky University, 2016
- Honors Results Based Merit Scholarship, University of Dhaka, 2010